

PA28 γ promotes the malignant progression of tumor by elevating mitochondrial function via C1QBP

Reviewed Preprint

v1 • October 1, 2024

Not revised

Jiongke Wang, Yujie Shi, Ying Wang, Yingqiang Shen, Huan Liu, Silu Sun, Yimei Wang, Xikun Zhou, Yu Zhou, Xin Zeng , Jing Li , Qianming Chen

State Key Laboratory of Oral Diseases & National Center for Stomatology & National Clinical Research Center for Oral Diseases & Research Unit of Oral Carcinogenesis and Management & Chinese Academy of Medical Sciences, West China Hospital of Stomatology, Sichuan University, Chengdu, China • Department of Stomatology, The First Affiliated Hospital with Nanjing Medical University, Nanjing, China • State Key Laboratory of Biotherapy and Cancer Center, West China Hospital, Sichuan University and Collaborative Innovation Center for Biotherapy, Chengdu, China

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Proteasome activator 28 γ (PA28 γ) plays a critical role in malignant progression of various tumors, however, its role and regulation are not well understood. Here, using oral squamous cell carcinoma (OSCC) as main research model, we discovered that PA28 γ interacted with complement 1q binding protein (C1QBP), which is dependent on the N-terminus of C1QBP rather than the known functional domain (amino acids 168-213). Notably, we found that PA28 γ can enhance C1QBP protein stability in OSCC. Functionally, PA28 γ contributes to the malignant progression of OSCC by affecting mitochondrial morphology and oxidative phosphorylation (OXPHOS) through C1QBP in vitro and vivo. Mechanically, PA28 γ upregulate the expression of optic atrophy 1 (OPA1), mitofusin 2 (MFN1), mitofusin 2 (MFN2) and the mitochondrial respiratory complex by C1QBP. Moreover, in a clinical cohort of OSCC patients, PA28 γ was positively correlated with C1QBP expression and negatively correlated with prognosis. Therefore, C1QBP is also a potential target for the treatment and prognosis of cancer.

eLife Assessment

This work attempts to demonstrate an ATP-independent non-canonical role of proteasomal component PA28 γ in the promotion of oral squamous cell carcinoma growth, migration, and invasion. The evidence around the following two areas remains **incomplete** and would benefit from further experimental work: 1) the stabilisation of the complement C1q binding protein (C1QBp) by PA28 γ , and 2) the impact of the PA28 γ -C1QBp interaction on mitochondrial function.

<https://doi.org/10.7554/eLife.101244.1.sa2>

Introduction

Proteasome activator 28γ (PA28γ), also known as REGγ or PSME3, is a well-known non-ATP-dependent proteasome regulator. PA28γ is overexpressed in some malignant tumors, such as thyroid, colon and breast cancer and is significantly associated with poor prognosis (Mao, Liu, Li, & Luo, 2008 [↗](#); Stadtmueller & Hill, 2011 [↗](#)). By regulating the stability of key components within various complex signaling pathways, PA28γ impacts the biological behavior of tumor cells. Our previous studies revealed that the expression level of PA28γ in oral squamous cell carcinoma (OSCC) cancer nest tissues is positively correlated with patient prognosis (J. Li et al., 2015 [↗](#); S. Liu et al., 2018 [↗](#)). However, its role and regulation are not well understood.

OSCC is the most common oral and maxillofacial malignant tumor (Siegel, Miller, Wagle, & Jemal, 2023 [↗](#); Sung et al., 2021 [↗](#)). The pathogenic factors of OSCC include smoking, drinking and viral infection, such as human papilloma virus (HPV). Therapies for OSCC include surgery, radiotherapy, chemotherapy and immunotherapy. Despite the rapid development of imaging, surgery, radiotherapy and immunotherapy in recent years, along with the emphasis on personalized treatment and multidisciplinary cooperation in the treatment of OSCC, the overall survival rate of OSCC patients has not significantly improved (Peres et al., 2019 [↗](#)). The main reasons for unsatisfactory treatment efficacy include invasion of surrounding tissue, early lymph node metastasis and a tendency toward recurrence (Lindemann, Takahashi, Patel, Osman, & Myers, 2018 [↗](#)). Metabolic reprogramming is a crucial characteristic of tumors, closely associated with malignant behaviors such as tumor growth, invasion, metastasis, and immune escape (Ohshima & Morii, 2021 [↗](#)). Current research on metabolic reprogramming in OSCC primarily focused on how fibroblasts regulate mitochondria to promote malignant tumor progression (X. Li et al., 2022 [↗](#); Zhang et al., 2020 [↗](#)). Understanding the molecular events that lead to the occurrence and development of OSCC can aid in developing new tumor-targeted therapies, which hold significant clinical application potential.

The complement 1q binding protein (C1QBP) is a crucial protein for maintaining mitochondrial function, particularly in mitochondrial oxidative phosphorylation (OXPHOS) (Ghebrehiwet, Geisbrecht, Xu, Savitt, & Peerschke, 2019 [↗](#); Q. Wang et al., 2022 [↗](#)). OXPHOS is essential for the production of adenosine triphosphate (ATP), which is necessary for tumor development. C1QBP has been shown to play a significant role in cancer progression, as influencing tumor growth, invasion and metastasis (Bai et al., 2019 [↗](#); Hou, Lu, Wang, & Yang, 2022 [↗](#); Vendramin et al., 2018 [↗](#)). Due to these roles, C1QBP presents a promising therapeutic target for various tumors, including melanoma, breast cancer, and colorectal cancer (Matsumoto & Bay, 2021 [↗](#)). In OSCC, enhanced mitochondrial OXPHOS function might also be linked to malignant tumor progression (Vyas et al., 2021 [↗](#); Xiao et al., 2021 [↗](#); Zhu, Liu, Wu, & Wang, 2021 [↗](#)), although the underlying mechanism remain unclear.

In this study, we discovered that PA28γ interacts with C1QBP and that PA28γ can stabilize C1QBP. This interaction enhances OXPHOS and promotes the growth, migration and invasion of OSCC cells. Additionally, the expression of PA28γ and C1QBP is increased and positively correlated in OSCC. Therefore, PA28γ and C1QBP are potential targets for the treatment and prognosis of cancer.

Results

PA28 γ interacts with and stabilizes C1QBP

To explore how PA28 γ promotes OSCC progression, we conducted a gene–gene interaction analysis using the GeneMANIA database. This analysis identified C1QBP as a significant gene within the interaction network, with the protein-coding gene of PA28 γ also playing a key role in the C1QBP interaction network (**Fig. 1A** and Appendix Fig. 1A). Subsequently, pull-down and coimmunoprecipitation analysis demonstrated that both endogenous and exogenous PA28 γ and C1QBP could bind to each other (**Fig. 1B, C** and Appendix Fig. 1B, C). Then, we developed four OSCC cell lines with stable overexpression of PA28 γ using a Flag-PA28 γ -expressing lentivirus and observed that the protein level of C1QBP was higher in these cell lines compared to control cells (**Fig. 1D**). Notably, PA28 γ upregulated C1QBP protein levels in a dose-dependent manner (**Fig. 1E**). To determine whether the upregulated C1QBP was due to increased stability, cells were pretreated with cycloheximide (CHX) to inhibit protein synthesis. This treatment minimally affected C1QBP degradation in the presence of PA28 γ (**Fig. 1F, G**). Furthermore, pull-down analysis revealed that the PA28 γ -C1QBP interaction depends on the N-terminus of C1QBP, rather than the known functional domain (amino acids 168-213) that bind to mitochondrial antiviral proteins (Peerschke et al., 2020; Xu, Xiao, Liu, Ren, & Gu, 2009) (**Fig. 1H, I**). Therefore, our results illustrated that PA28 γ could interact with C1QBP, in a manner dependent on the N-terminus of C1QBP, and that PA28 γ could stabilize C1QBP.

PA28 γ and C1QBP colocalize in mitochondria and affect mitochondrial functions

To further investigate the physical interaction between PA28 γ and C1QBP, we conducted an immunofluorescence (IF) assay in two OSCC cell lines, as shown these two proteins were colocalized in the mitochondria (**Fig. 2A**). Remarkably, transmission electron microscopy (TEM) images revealed fewer mitochondrial vacuoles and higher mitochondrial ridge density in PA28 γ -overexpressing cells compared to control cells (**Fig. 2B** and Appendix Fig. 2A). IF analysis also showed increased mitochondrial lengths and areas in the PA28 γ -overexpressing cells (**Fig. 2C, D**). Following that, we conducted a Cell Mito Stress Test to measure the oxygen consumption rate (OCR) (**Fig. 2E, F**). The results indicated significantly higher basal respiration, maximal OCRs and ATP production were significantly greater in PA28 γ -overexpressing cells compared to control cells (**Fig. 2G-I** and Appendix Fig. 2B-D). Considering that mitochondria are the primary site for ROS generation and that ROS play a crucial role in carcinogenesis, we measured ROS levels in OSCC cells. Consistently, ROS levels were significantly higher in PA28 γ -overexpressing cells (**Fig. 2J** and Appendix Fig. 2E). In addition, experiments conducted in PA28 γ -sh OSCC cells yielded opposite results, further confirming these conclusions (Appendix Fig. 2F-K).

To demonstrate the carcinogenic abilities of PA28 γ and C1QBP, we injected 4MOSC2 cells, a mouse OSCC line, with or without Pa28 γ overexpression under the surface of the tongue in C57BL/6 mice (Appendix **Fig. 3A**). After 10 days of normal feeding, the tumors in the Pa28 γ -overexpressing groups were significantly larger than those in the control group. Furthermore, via immunohistochemical (IHC) staining revealed that that PA28 γ overexpression upregulate C1QBP in vivo (**Fig. 3A-C**). Consistent with the in vitro experiments, ATP production and ROS generation were significantly increased (**Fig. 3D, E**). Additionally, analysis of OSCC xenograft tumor tissue in nude mice revealed that PA28 γ 's regulation of C1QBP in OSCC cells is independent of the immune system (Appendix Fig. 3B, C). In addition, we established stable Pa28 γ -silenced B16 cells, a mouse melanoma cell line, and found that the level of C1qbp protein was decreased in Pa28 γ -silenced B16 cells (Appendix Fig. 3D). Cells with or without Pa28 γ silencing were grafted into the flanks of Pa28 γ ^{ko/ko} and Pa28 γ ^{wt/wt} C57BL/6 mice by subcutaneous injection (Appendix **Fig. 3E**). Remarkably, the tumors derived from the silenced Pa28 γ cells in Pa28 γ ^{ko/ko} mice were

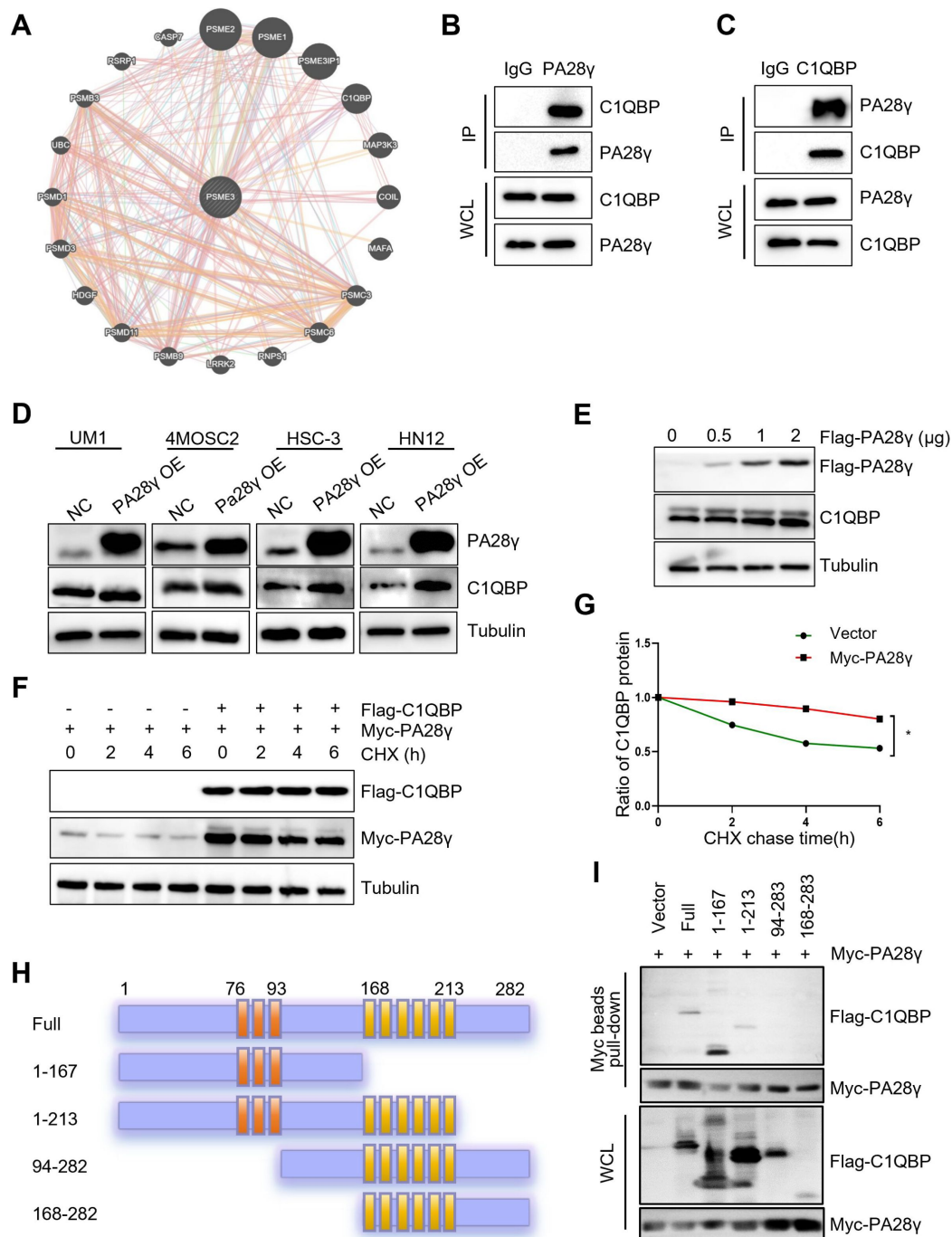


Fig. 1

The interaction between PA28γ and C1QBP.

(A) The gene network of PA28γ and C1QBP in the GeneMANIA database. (B, C) The interaction between endogenous PA28γ and C1QBP in HSC-3 cells was verified via immunoprecipitation. (D) Western blot analysis of C1QBP in four OSCC cell lines with PA28γ-overexpressing. (E) Western blot analysis of C1QBP in 293T cells transfected with increasing doses of Flag-PA28γ. (F) 293T cells transfected with Flag-C1QBP with or without Myc-PA28γ were treated with CHX (100 μg/ml) for the indicated periods of time. (G) Quantification of Flag-C1QBP levels relative to tubulin levels is shown (the data are representative of 1 experiment with 3 independent biological replicates, * $P < 0.05$). (H) Full-length C1QBP and truncation with deletion of functional domains. (I) Pull-down of 293T cells transfected with Myc-PA28γ and full-length Flag-C1QBP or truncation mutants of functional domains for 36h.

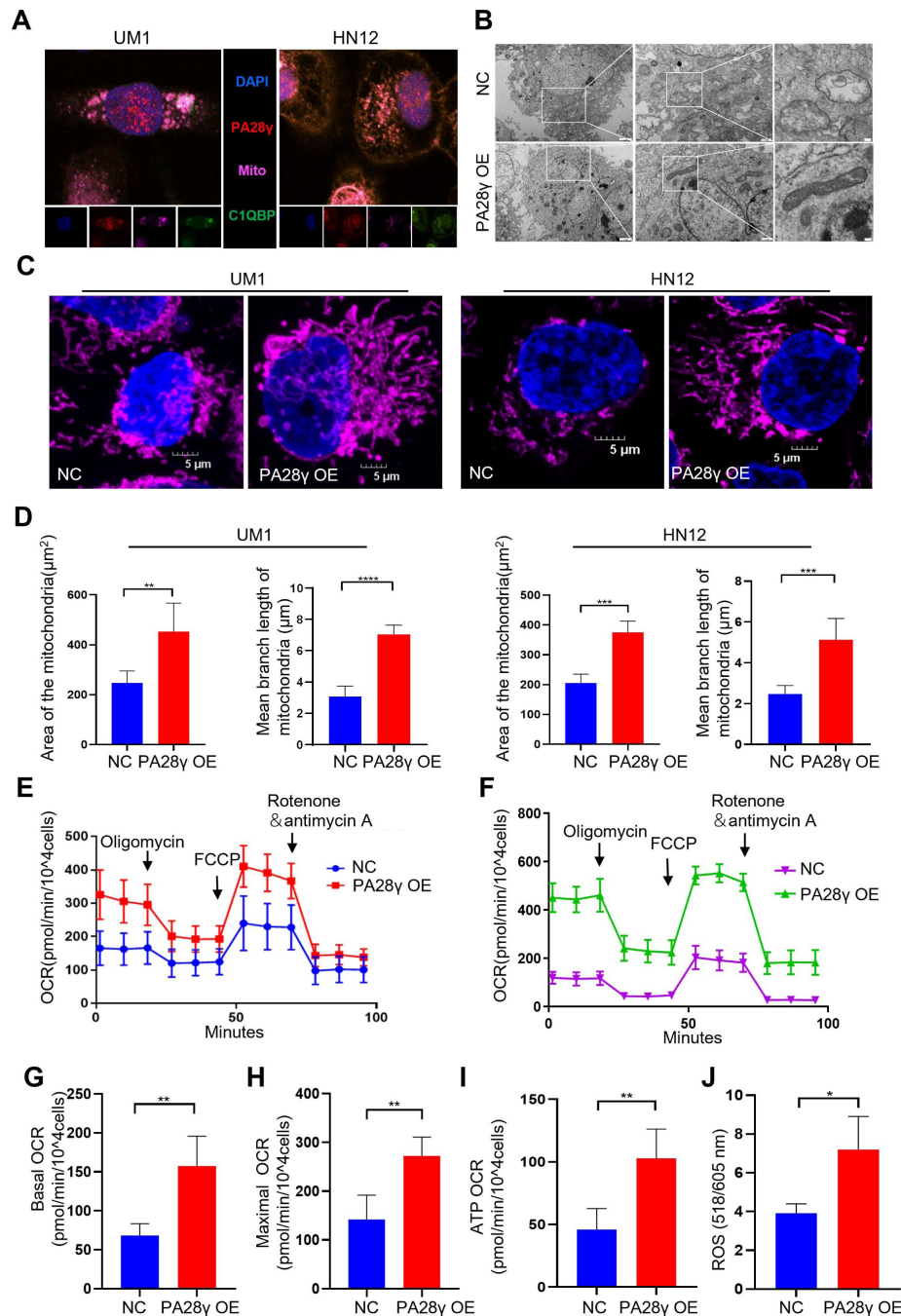


Fig. 2

PA28γ and C1QBP colocalize in mitochondria and affect mitochondrial functions *in vitro*.

(A) Confocal image of IF in two OSCC cell lines. (B) TEM images of PA28γ-overexpressing and control UM1 cells. (C) Representative confocal images of mitochondria in two OSCC cell lines. (D) The area and mean branch length of mitochondria in two OSCC cell lines were measured by ImageJ (the data are presented as the mean ± SD of 3 independent experiments; **P<0.01, ***P<0.001, and ****P<0.0001). (E, F) OCRs of PA28γ-overexpressing and control UM1 and HN12 cells were plotted using a Cell Mito Stress Test Kit (the data are presented as the means ± SDs of 3 independent experiments). (G-I) Basal OCRs, maximal OCRs and ATP production of PA28γ-overexpressing and control UM1 cells measured by the Cell Mito Stress Test (the data are presented as the means ± SDs of 3 independent experiments, **P<0.01). (J) ROS generation in PA28γ-overexpressing and control UM1 cells (the data are presented as the means ± SDs of 3 independent experiments; *P<0.05).

smaller than those in the other groups (Fig. 3F-H). The trend of ATP and ROS was consistent with tumor size (Fig. 3I, J), and the protein levels of C1qbp in the tumors were also reduced in the PA28γ-silenced groups (Fig. 3K).

Collectively, these data suggest that PA28γ, which co-localizes with C1QBP in mitochondria, is involved in regulating mitochondrial morphology and function.

PA28γ regulates mitochondrial OXPHOS by upregulating C1QBP

Next, we explored the protein levels of the mitochondrial respiratory chain complex and other mitochondrial functional proteins. We found that the levels of complex I and

IV proteins, as well as OPA1, MFN1 and MFN2 proteins, were upregulated in PA28γ-overexpressing OSCC cells (Fig. 4A, B and Appendix Fig. 4A, B). These increases could be reversed by C1QBP silencing in PA28γ-overexpressing OSCC cells (Fig. 4C).

Consistent with this, the OCRs of C1QBP-silenced PA28γ-overexpressing OSCC cells were markedly lower than those of control cells (Fig. 4D and Appendix Fig. 4C-E). Subsequently, we constructed a series of experiments to detect the biological behavior of control and PA28γ-overexpressing OSCC cells with or without C1QBP silencing. C1QBP knockdown significantly attenuated the migration, invasive and proliferation capabilities previously augmented by PA28γ overexpression (Fig. 4E-H). These data suggest that PA28γ enhances mitochondrial OXPHOS function through C1QBP.

C1QBP expression was positively associated with PA28γ expression and was associated with a worse prognosis in patients with OSCC or SKCM

To determine the correlation between PA28γ and C1QBP, IHC staining was performed in the clinical cohort of oral mucosa carcinogenesis. The results showed that both C1QBP and PA28γ were notably upregulated during the progression of oral mucosa carcinogenesis (Fig. 5A, B and Appendix Fig. 5A), and the levels of PA28γ and C1QBP were positively related (Fig. 5C). Similarly, both C1QBP and PA28γ were upregulated in metastatic OSCC tissues (Fig. 5D, E and Appendix Fig. 5B), and the levels of PA28γ and C1QBP were consistently positively correlated in OSCC cohort (Fig. 5F). Notably, consistent with our previous findings on PA28γ (J. Li et al., 2015), C1QBP or combining PA28γ and C1QBP could be a negative predictor in our multicenter OSCC clinical cohort (Fig. 5G, H), TCGA HNSC database (Fig. 5I, J) and TCGA SKCM database (Appendix Fig. 5C, D).

Discussion

Metabolic reprogramming is one of the hallmarks of malignant tumors and is related to the malignant biological behavior of tumors (Faubert, Solmonson, & DeBerardinis, 2020; Hanahan, 2022; Tsai et al., 2023; Xia et al., 2021). The metabolic phenotype of tumor cells not only differs from that of normal cells but also dynamically changes during tumor progression. OSCC is a common head and neck malignancy that is still a growing global health problem (Johnson et al., 2020). The development of OSCC is a complex and multifaceted process in which metabolic reprogramming appears to be important (B. Liu, Si, Wei, Zhang, & Chen, 2023). However, the mechanism of metabolic reprogramming remains unclear. Excitingly, we found solid evidence that PA28γ interacts with and stabilizes C1QBP. Aberrantly accumulated C1QBP enhances the function of mitochondrial OXPHOS and leads to the production of additional ATP and ROS by activating the expression and function of OPA1, MNF1, MFN2 and mitochondrial respiratory chain complex

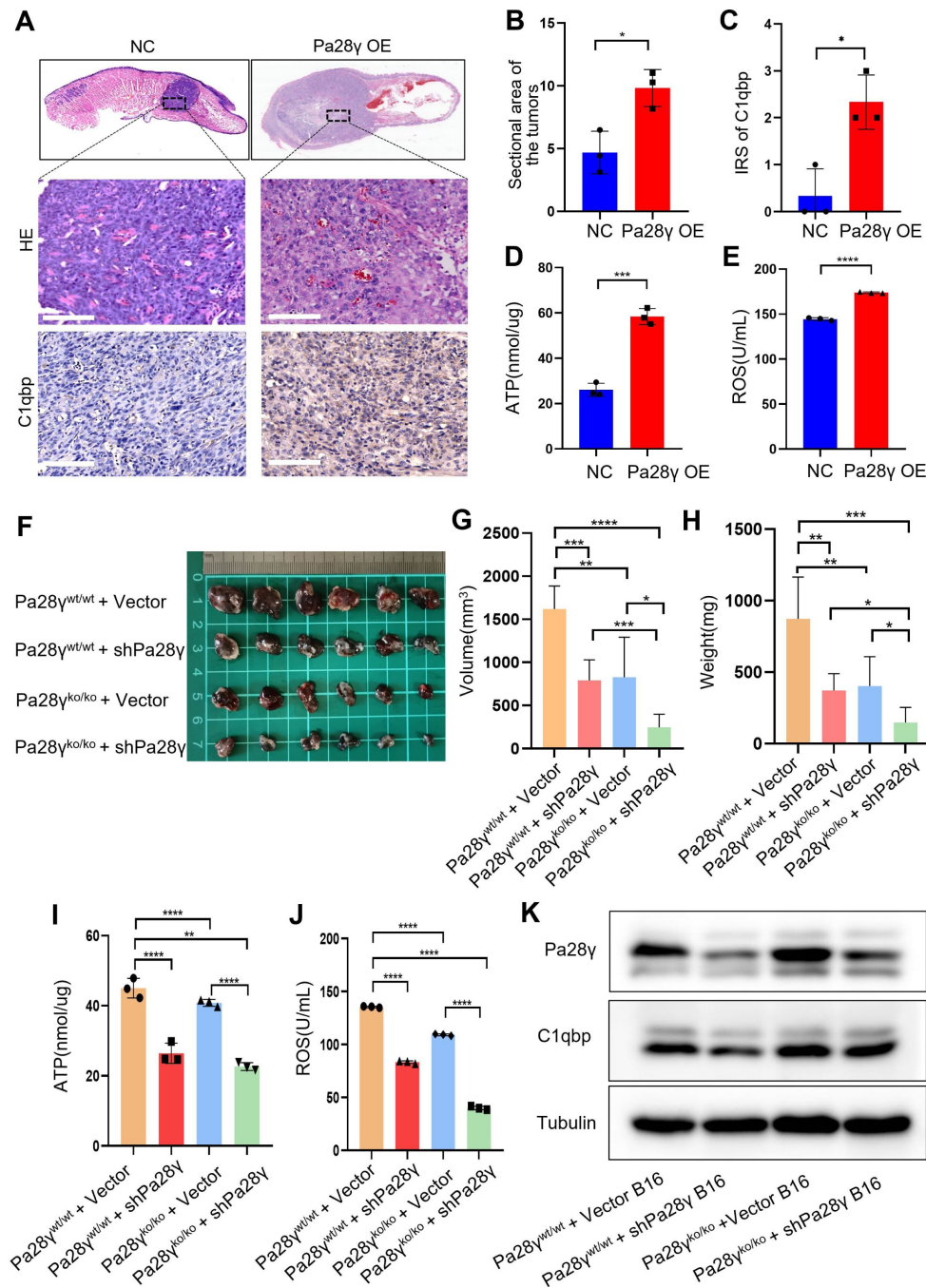


Fig. 3

Pa28y affect mitochondrial functions *in vivo*.

(A) Representative images of H&E staining and IHC staining of C1qbp of tongue sections from mice (n=3). **(B)** Quantification of the sectional area of tumors from the Pa28y-overexpressing and control groups (n=3; the data are presented as the means $\pm$ SDs of 3 samples; $*P < 0.05$). **(C)** Comparison of the immunoreactive scores (IRSs) of C1qbp antibody staining in the Pa28y-overexpressing and control groups (n=3; the data are presented as the means $\pm$ SDs; $*P < 0.05$). **(D, E)** Quantification of ATP production and ROS levels in tumors from the Pa28y-overexpressing and control groups (n=3; the data are presented as the means $\pm$ SDs of 3 samples; $*P < 0.05$, $**P < 0.001$, $***P < 0.0001$). **(F)** Images of the tumors in different groups at the endpoint (n=6). **(G-J)** The volume, weight, ATP and ROS of tumors in different groups at the endpoint (n=6; the data are presented as the means $\pm$ SDs of 3 independent experiments; $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, $****P < 0.0001$). **(K)** Western blot analysis of tumors in different groups.

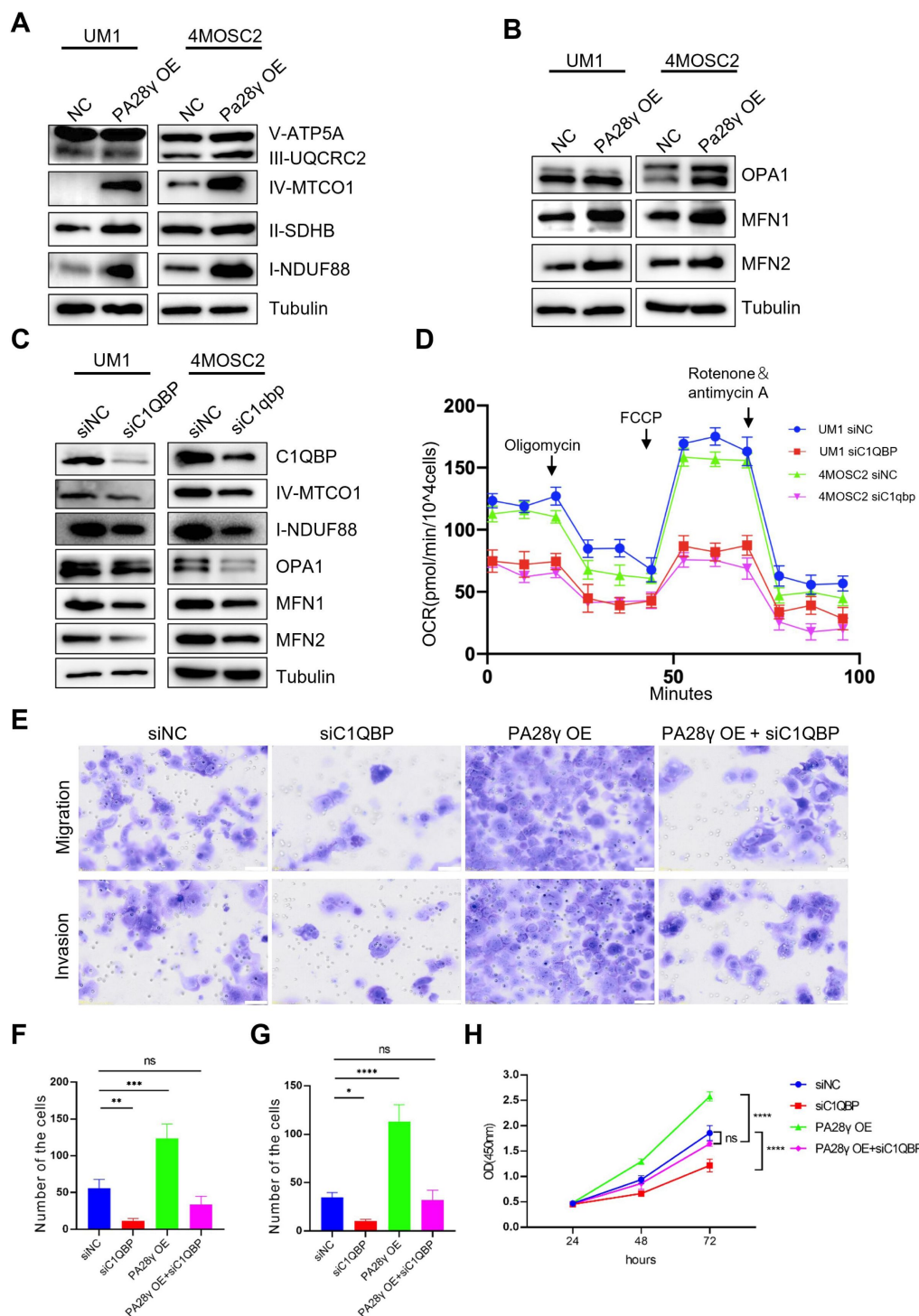


Fig. 4

PA28γ regulates mitochondrial OXPHOS and cellular biological behavior via C1QBP.

(A, B) Western blot analysis of PA28γ-overexpressing and control UM1 and 4MOSC2 cells. (C) Western blot analysis of PA28γ-overexpressing OSCC cells transfected with siNC or siC1QBP. (D) OCRs of C1QBP-silenced and control PA28γ-overexpressing UM1 and 4MOSC2 cells (the data are presented as the means ± SDs of 3 independent experiments). (E-H) Cell migration, invasion and proliferation in control, C1QBP-silenced, PA28γ-overexpressing and PA28γ-overexpressing + C1QBP-silenced UM1 cells (the data are presented as the means ± SDs of 3 independent experiments; *P<0.05, **P<0.01, ***P<0.001, ****P<0.0001).

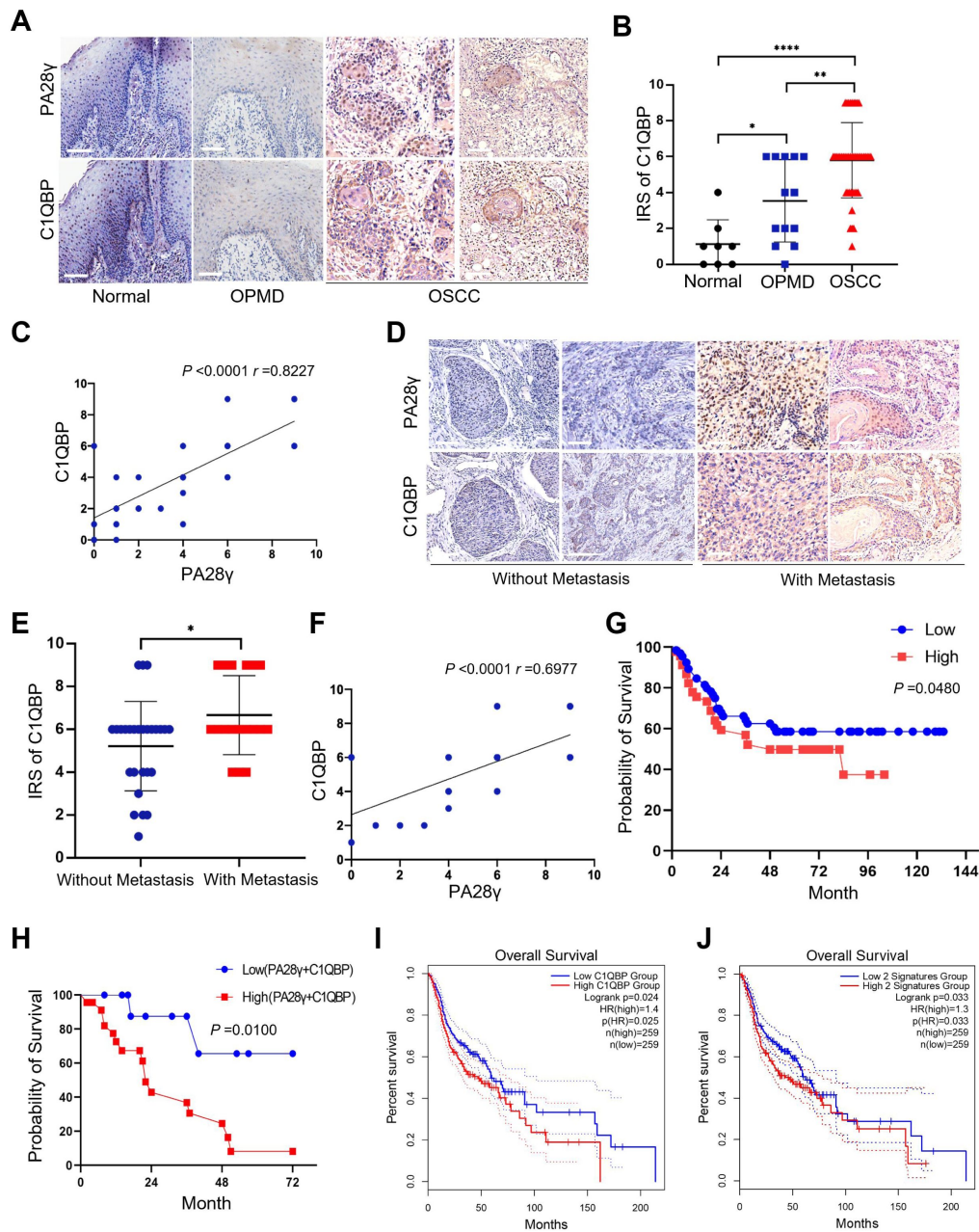


Fig. 5

The correlation between PA28y and C1QBP in the carcinogenesis and development of OSCC.

(A) Representative IHC staining of PA28y and C1QBP in normal ($n=8$), OPMD ($n=13$) and OSCC ($n=45$) samples. **(B)** Comparison of the immunoreactive scores (IRSs) of C1QBP between the normal, OPMD and OSCC groups (the data are presented as the means $\pm$ SDs; $*P<0.05$, $**P<0.01$, $***P<0.0001$). **(C)** Spearman correlation analysis was used to test the correlation between PA28y and C1QBP in normal, OPMD and OSCC tissues ($P<0.0001$, $r=0.8227$). **(D)** Representative IHC staining of PA28y and C1QBP in nonmetastatic ($n=27$) and metastatic ($n=18$) OSCC patients. **(E)** Comparison of the IRSs of C1QBP in the nonmetastatic and metastatic OSCC groups (the data are presented as the means $\pm$ SDs; $*P<0.05$). **(F)** Spearman correlation analysis was used to test the correlation between PA28y and C1QBP in OSCC tissues ($P<0.0001$, $r=0.6977$). **(G)** Kaplan–Meier analysis of the protein expression of C1QBP in our multicenter OSCC clinical cohort ($n=295$, $P=0.0480$). **(H)** Kaplan–Meier analysis of both low or high protein expression of C1QBP and PA28y in our multicenter OSCC clinical cohort ($n=295$, $P=0.0100$). **(I)** Kaplan–Meier analysis of the protein expression of C1QBP in TCGA HNSC database ($n=259$, $P=0.025$). **(J)** Kaplan–Meier analysis of both low or high protein expression of C1QBP and PA28y in TCGA HNSC database ($n=259$, $P=0.033$).

proteins. This process results in resulting in mitochondrial fusion and malignant tumor progression (**Fig. 6**). Our study highlights the underlying mechanism by which PA28 γ participates in the regulation of OXPHOS by upregulating C1QBP in OSCC.

C1QBP is also called gClqR, p32, p33 and HABP1, can be distributed inside cells, on the cell surface, and can also be secreted extracellularly (Egusquiza-Alvarez & Robles-Flores, 2022). Within cells, C1QBP is primarily located in the mitochondria. The key functions of mitochondrial C1QBP include maintaining OXPHOS, supplying energy to cells, and achieving homeostasis of mitochondrial metabolism (Ghebrehiwet et al., 2019; Q. Wang et al., 2022). C1QBP is highly expressed in cells that require a large amount of energy, such as the heart, skeletal muscle, testis, ovary, small intestine and colon (Feichtinger et al., 2017). Sufficient energy is essential for cancer development, and C1QBP promotes malignant behaviors such as tumor invasion and metastasis by enhancing OXPHOS function in various cancers (Raschdorf et al., 2021; Q. Wang et al., 2022). However, there are no reports on the expression and functions of C1QBP in OSCC. In our study, we found solid evidence that PA28 γ interacts with and stabilizes C1QBP. As a proteasome activator, PA28 γ usually affects the development of tumors by regulating the stability of various important proteins. For example, PA28 γ can promote the degradation of p53, p21 leading to cancer progression (J. Liu et al., 2010). However, the function of mitochondrial PA28 γ may be differ from that of nuclear PA28 γ (Mao et al., 2008). PA28 γ interacts with C1QBP and stabilizes C1QBP at the protein level, suggesting that the binding sites of PA28 γ and C1QBP may mask the specific post-translational modification sites of C1QBP and inhibit its degradation. For example, the transcriptional coactivator p300 interacts with Smad7 by acetylating two lysine residues in its N-terminus, which stabilize Smad7 and protect it from TGF β -induced degradation (Grönroos, Hellman, Heldin, & Ericsson, 2002). Overall, the specific mechanism involved in the interaction between PA28 γ and C1QBP still needs further exploration.

Considering that C1QBP is a vital protein for maintaining mitochondrial metabolism (Tian et al., 2023), we further investigated the function underlying the malignant progression of OSCC through the interaction between PA28 γ and C1QBP in vitro and in vivo. PA28 γ and C1QBP colocalized in the mitochondria, and the stabilization of C1QBP by PA28 γ enhanced mitochondrial morphology and OXPHOS. This enhancement is crucial for ATP production, and ATP is the primary energy currency of cellular metabolism. Therefore, PA28 γ 's regulation of OXPHOS directly impacts cellular energy metabolism. PA28 γ has been reported to activate the mTORC1 signaling pathway in hepatocellular carcinoma cells to promote glycolysis and inhibit OXPHOS (Yao et al., 2021). These findings contrast with our study, potentially due to organ heterogeneity or the complexity of metabolic reprogramming.

In the tissue microenvironment of oral lichen planus, a potentially malignant oral disorder, PA28 γ in epithelial cells can regulate T cell differentiation (Y. Wang et al., 2024), while PA28 γ in CAFs is involved in the crosstalk between stromal cells and tumor cells (Z. Li et al., 2024). This suggests that PA28 γ may interact with the tumor immune microenvironment. Our phenotypically similar OSCC xenograft tumor models in both immunocompetent and nude mice indicate that the regulation of mitochondrial OXPHOS by PA28 γ localized in tumor cell mitochondria does not entirely depend on the immune system (Z. Wang et al., 2019).

Furthermore, we confirmed that PA28 γ regulates C1QBP and influences mitochondrial morphology and function by enhancing the expression of OPA1, MFN1, MFN2 and the mitochondrial respiratory complex. Mitochondrial fusion, crucial for oxidative metabolism and cell proliferation, is regulated by the large GTPases MFN1 and MFN2, which fuse with the outer mitochondrial membrane, and OPA1, which fuses with the inner mitochondrial membrane (Westermann, 2010). OPA1 is an essential inner mitochondrial membrane protein with multiple functions, including regulating mitochondrial fusion, crest morphology, mtDNA stability, and interacting with the mitochondrial respiratory chain to regulate OXPHOS (Del Dotto et al., 2018).

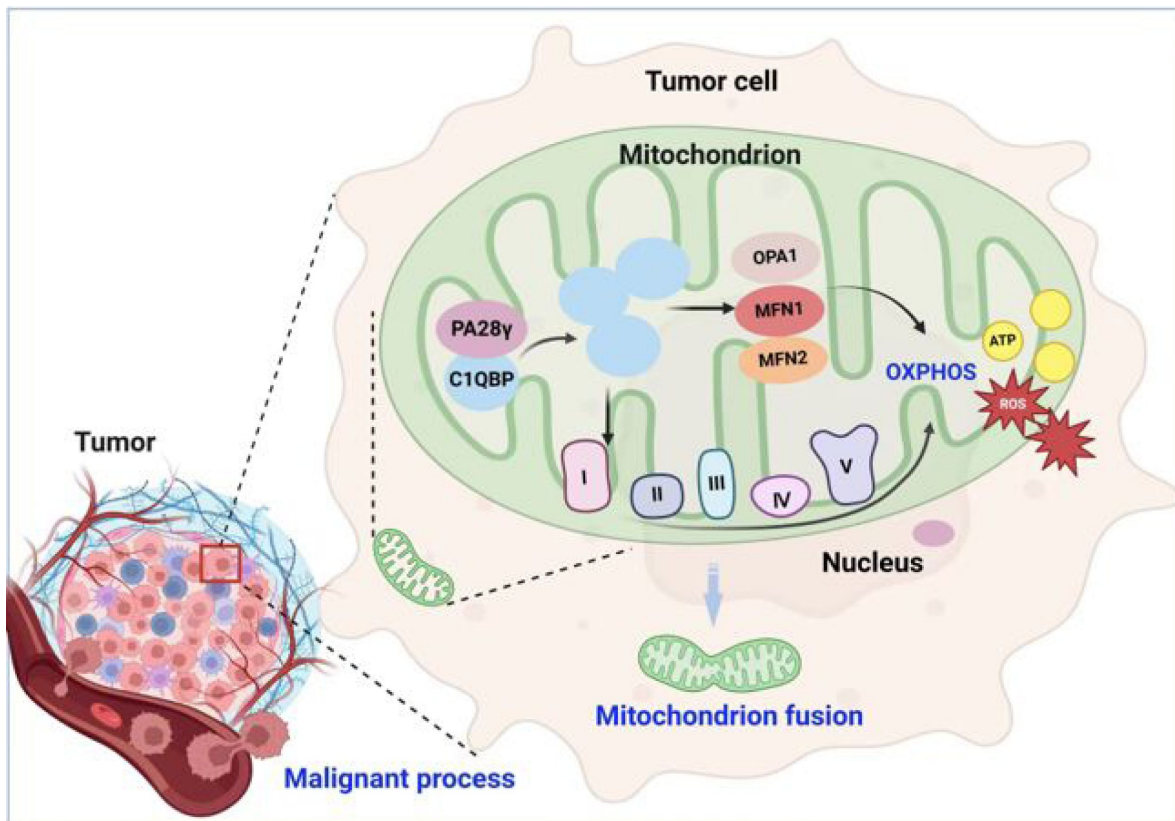


Fig. 6

Molecular mechanism through which PA28γ interacts with C1QBP in the malignant progression of tumor.

PA28γ can interact with and stabilize C1QBP, which can activate the expression and function of OPA1, MFN1, MFN2 and mitochondrial respiratory chain complex proteins, resulting in enhanced mitochondrial OXPHOS, mitochondrial fusion and malignant tumor progression.

Finally, we analyzed the expression patterns and significance of PA28 γ and C1QBP in a clinical cohort. PA28 γ and C1QBP were negatively correlated with prognosis in the OSCC clinical cohort. Notably, some studies have shown that OXPHOS is upregulated in various tumors, with some advanced tumors preferring OXPHOS metabolically (Ohshima & Morij, 2021 [DOI](#); Qiu, Li, & Zhang, 2023 [DOI](#)). OXPHOS activity is associated with the recurrence of OSCC (Noh et al., 2023 [DOI](#)). Additionally, the enhancement of OXPHOS and ATP production by ROS proto-oncogene 1, which localizes to mitochondria in OSCC, promotes the OSCC invasion (Noh et al., 2023 [DOI](#); Grimm et al., 2014 [DOI](#)). These findings are consistent with our results. Our study not only explain the mechanisms and signaling networks underlying the oncogenic potential of PA28 γ but also contribute to understanding the possible mechanisms involved in OSCC metabolic reprogramming.

In summary, we found that PA28 γ could interact with C1QBP and stabilize C1QBP. This enhances the function of OXPHOS, leading to increased production of ATP and ROS by activating the expression and function of OPA1, MNF1, MFN2 and the mitochondrial respiratory chain complex. This process promotes mitochondrial fusion and malignant tumor progression. The expression of PA28 γ and C1QBP was increased and positively correlated in OSCC. High expression of PA28 γ and C1QBP was correlated with poor prognosis in OSCC patients, indicating that both factors could serve as potential targets for the treatment and prognosis of OSCC.

Materials and methods

Patients and Follow-up

Our study included 4 cohorts. The first cohort included 8 normal controls, 13 patients with oral potentially malignant disorder (OPMD), and 45 patients with OSCC (Appendix Table 1). These patients visited the West China Hospital of Stomatology in 2021. The normal tissues were obtained from patients who underwent maxillofacial plastic surgery. The second cohort for survival analysis included 295 patients diagnosed with primary OSCC tumors who visited West China Hospital of Stomatology, Peking University Hospital of Stomatology and Guangdong Provincial Stomatological Hospital from 2005 to 2009 (Appendix Table 2). None of the patients had cancer in other organs. These patients received regular follow-up for 1-133 months after the operation. The protocol of this study was approved by the ethics committee of the West China Hospital of Stomatology (Approval number: WCHSIRB-D-2020-046). The third cohort, comprising 518 patients with head and neck squamous cell carcinoma (HNSCC), was obtained from The Cancer Genome Atlas (TCGA) database (Appendix Table 3). The last cohort, comprising 458 patients with skin cutaneous melanoma (SKCM), was obtained from TCGA database (Appendix Table 4).

Laboratory Experiments

All animal studies were approved by the Animal Care and Use Committee for the State Key Laboratory of Oral Diseases (Approval number: WCHSIRB-D-2022-032) in compliance with the Guide for the updated Animal Research: Reporting of In Vivo Experiments (ARRIVE) 2.0 guidelines. Other methods are detailed in the *Supplemental Materials and Methods*, including cell culture and cell-related experiments, animal models, mitochondrial function detection, western blot/immunoprecipitation assays, hematoxylin and eosin (H&E) staining, and immunohistochemistry.

Statistical Analysis

The data were analyzed using GraphPad Prism software version 8.0. The statistical data are expressed as the mean $\pm$ standard deviation (SD). Chi-square and Fisher exact tests, Spearman correlation tests, Wilcoxon rank-sum tests, one-way ANOVA and Student's t tests were used to analyze the data. The Kaplan–Meier method was used for survival analysis, and the log-rank test

was used to evaluate the prognostic value of C1QBP in OSCC patients. The data are shown as the mean $\pm$ standard error, and a value of $P < 0.05$ was considered to indicate statistical significance: $P < 0.05^*$, $P < 0.01^{**}$, $P < 0.001^{***}$, and $P < 0.0001^{****}$. Each experiment was repeated 3 times.

Acknowledgements

This work was supported by grants from the National Natural Science Foundation of China (82072999 and 82273320 to J. Li, U19A2005 and 82270986 to X. Zeng, 82201074 to J. Wang), the Province Natural Science Foundation of Sichuan (23NSFSC0145 to J. Li), and the Innovation Research Project of Sichuan University (2022SCUH0029 to J. Li).

Author contributions

J. Wang, Y. Shi, X. Zeng, J. Li contributed to conception and design, data acquisition, analysis, and interpretation, drafted and critically revised the manuscript; Y. Wang, Y. Shen, H. Liu contributed to data acquisition; S. Sun, Y. Wang contributed to data analysis and interpretation; X. Zhou, Q. Chen contributed to conception and design, critically revised the manuscript. All the authors gave final approval and agreed to be accountable for all aspects of the work.

Conflict of interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

References

- Bai Y., Wang W., Li S., Zhan J., Li H., Zhao M., Wang J. (2019) **C1QBP Promotes Homologous Recombination by Stabilizing MRE11 and Controlling the Assembly and Activation of MRE11/RAD50/NBS1 Complex** *Mol Cell* **75**:1299–1314 <https://doi.org/10.1016/j.molcel.2019.06.023>
- Del Dotto V., Fogazza M., Lenaers G., Rugolo M., Carelli V., Zanna C (2018) **OPA1: How much do we know to approach therapy?** *Pharmacol Res* **131**:199–210 <https://doi.org/10.1016/j.phrs.2018.02.018>
- Egusquiza-Alvarez C. A., Robles-Flores M (2022) **An approach to p32/gC1qR/HABP1: a multifunctional protein with an essential role in cancer** *J Cancer Res Clin Oncol* **148**:1831–1854 <https://doi.org/10.1007/s00432-022-04001-5>
- Faubert B., Solmonson A., DeBerardinis R. J (2020) **Metabolic reprogramming and cancer progression** *Science* **368** <https://doi.org/10.1126/science.aaw5473>
- Feichtinger R. G., Oláhová M., Kishita Y., Garone C., Kremer L. S., Yagi M., Prokisch H. (2017) **Biallelic C1QBP Mutations Cause Severe Neonatal-, Childhood-, or Later-Onset Cardiomyopathy Associated with Combined Respiratory-Chain Deficiencies** *The American Journal of Human Genetics* **101**:525–538 <https://doi.org/10.1016/j.ajhg.2017.08.015>
- Ghebrehiwet B., Geisbrecht B. V., Xu X., Savitt A. G., Peerschke E. I. B (2019) **The C1q Receptors: Focus on gC1qR/p33 (C1qBP, p32, HABP-1)** *Semin Immunol* **45** <https://doi.org/10.1016/j.smim.2019.101338>
- Grimm M., Cetindis M., Lehmann M., Biegner T., Munz A., Teriete P., Reinert S. (2014) **Association of cancer metabolism-related proteins with oral carcinogenesis – indications for chemoprevention and metabolic sensitizing of oral squamous cell carcinoma?** *Journal of Translational Medicine* **12** <https://doi.org/10.1186/1479-5876-12-208>
- Grönroos E., Hellman U., Heldin C. H., Ericsson J (2002) **Control of Smad7 stability by competition between acetylation and ubiquitination** *Mol Cell* **10**:483–493 [https://doi.org/10.1016/s1097-2765\(02\)00639-1](https://doi.org/10.1016/s1097-2765(02)00639-1)
- Hanahan D (2022) **Hallmarks of Cancer: New Dimensions** *Cancer Discov* **12**:31–46 <https://doi.org/10.1158/2159-8290.cd-21-1059>
- Hou G., Lu Z., Wang Z., Yang X (2022) **The Mitochondrial Protein C1QBP Promotes Hepatocellular Carcinoma Progression by Enhancing Cell Survival, Migration and Invasion** *J Cancer* **13**:2477–2489 <https://doi.org/10.7150/jca.69379>
- Johnson D. E., Burtneß B., Leemans C. R., Lui V. W. Y., Bauman J. E., Grandis J. R (2020) **Head and neck squamous cell carcinoma** *Nat Rev Dis Primers* **6** <https://doi.org/10.1038/s41572-020-00224-3>

- Li J., Feng X., Sun C., Zeng X., Xie L., Xu H., Chen Q. (2015) **Associations between proteasomal activator PA28y and outcome of oral squamous cell carcinoma: Evidence from cohort studies and functional analyses** *EBioMedicine* **2**:851–858 <https://doi.org/10.1016/j.ebiom.2015.07.004>
- Li X., Jiang E., Zhao H., Chen Y., Xu Y., Feng C., Shang Z. (2022) **Glycometabolic reprogramming-mediated proangiogenic phenotype enhancement of cancer-associated fibroblasts in oral squamous cell carcinoma: role of PGC-1α/PFKFB3 axis** *Br J Cancer* **127**:449–461 <https://doi.org/10.1038/s41416-022-01818-2>
- Li Z., Sun S., Wang Y., Hua Y., Liu M., Zhou Y., Li J. (2024) **PA28y coordinates the cross-talk between cancer-associated fibroblasts and tumor cells to promote OSCC progression via HDAC1/E2F3/IGF2 signaling** *Cancer Lett* **594** <https://doi.org/10.1016/j.canlet.2024.216962>
- Lindemann A., Takahashi H., Patel A. A., Osman A. A., Myers J. N (2018) **Targeting the DNA Damage Response in OSCC with TP53 Mutations** *J Dent Res* **97**:635–644 <https://doi.org/10.1177/0022034518759068>
- Liu B., Si W., Wei B., Zhang X., Chen P (2023) **PTP4A1 promotes oral squamous cell carcinoma (OSCC) metastasis through altered mitochondrial metabolic reprogramming** *Cell Death Discov* **9** <https://doi.org/10.1038/s41420-023-01657-x>
- Liu J., Yu G., Zhao Y., Zhao D., Wang Y., Wang L., Li X. (2010) **REGgamma modulates p53 activity by regulating its cellular localization** *J Cell Sci* **123**:4076–4084 <https://doi.org/10.1242/jcs.067405>
- Liu S., Liu D., Zeng X., Wang J., Liu J., Cheng J., Chen Q. (2018) **PA28y acts as a dual regulator of IL-6 and CCL2 and contributes to tumor angiogenesis in oral squamous cell carcinoma** *Cancer Lett* **428**:192–200 <https://doi.org/10.1016/j.canlet.2018.04.024>
- Mao I., Liu J., Li X., Luo H (2008) **REGgamma, a proteasome activator and beyond?** *Cell Mol Life Sci* **65**:3971–3980 <https://doi.org/10.1007/s00018-008-8291-z>
- Matsumoto K., Bay B. H (2021) **Role of C1QBP/p32 and its Therapeutic Potential in Breast Carcinoma and other Cancers** *Curr Med Chem* **28**:5048–5065 <https://doi.org/10.2174/0929867328666201231124038>
- Noh J. K., Woo S. R., Kong M., Lee M. K., Lee J. W., Lee Y. C., Eun Y. G (2023) **Gene signature predicting recurrence in oral squamous cell carcinoma is characterized by increased oxidative phosphorylation** *Mol Oncol* **17**:134–149 <https://doi.org/10.1002/1878-0261.13328>
- Ohshima K., Morii E (2021) **Metabolic Reprogramming of Cancer Cells during Tumor Progression and Metastasis** *Metabolites* **11** <https://doi.org/10.3390/metabo11010028>
- Peerschke E., Stier K., Li X., Kandov E., de Stanchina E., Chang Q., Adusumilli P. S. (2020) **gC1qR/HABP1/p32 Is a Potential New Therapeutic Target Against Mesothelioma** *Front Oncol* **10** <https://doi.org/10.3389/fonc.2020.01413>
- Peres M. A., Macpherson L. M. D., Weyant R. J., Daly B., Venturelli R., Mathur M. R., Watt R. G (2019) **Oral diseases: a global public health challenge** *Lancet* **394**:249–260 [https://doi.org/10.1016/s0140-6736\(19\)31146-8](https://doi.org/10.1016/s0140-6736(19)31146-8)

- Qiu X., Li Y., Zhang Z (2023) **Crosstalk between oxidative phosphorylation and immune escape in cancer: a new concept of therapeutic targets selection** *Cell Oncol (Dordr)* **46**:847–865 <https://doi.org/10.1007/s13402-023-00801-0>
- Raschdorf A., Sünderhauf A., Skibbe K., Ghebrehiwet B., Peerschke E. I., Sina C., Derer S (2021) **Heterozygous P32/C1QBP/HABP1 Polymorphism rs56014026 Reduces Mitochondrial Oxidative Phosphorylation and Is Expressed in Low-grade Colorectal Carcinomas** *Frontiers in Oncology* **10** <https://doi.org/10.3389/fonc.2020.631592>
- Siegel R. L., Miller K. D., Wagle N. S., Jemal A (2023) **Cancer statistics, 2023** *CA Cancer J Clin* **73**:17–48 <https://doi.org/10.3322/caac.21763>
- Stadtmueller B. M., Hill C. P (2011) **Proteasome Activators** *Molecular Cell* **41**:8–19 <https://doi.org/10.1016/j.molcel.2010.12.020>
- Sung H., Ferlay J., Siegel R. L., Laversanne M., Soerjomataram I., Jemal A., Bray F (2021) **Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries** *CA Cancer J Clin* **71**:209–249 <https://doi.org/10.3322/caac.21660>
- Tian H., Chai D., Wang G., Wang Q., Sun N., Jiang G., Zheng J. (2023) **Mitochondrial C1QBP is essential for T cell antitumor function by maintaining mitochondrial plasticity and metabolic fitness** *Cancer Immunol Immunother* **72**:2151–2168 <https://doi.org/10.1007/s00262-023-03407-5>
- Tsai C. H., Chuang Y. M., Li X., Yu Y. R., Tzeng S. F., Teoh S. T., Ho P. C (2023) **Immunoediting instructs tumor metabolic reprogramming to support immune evasion** *Cell Metab* **35**:118–133 <https://doi.org/10.1016/j.cmet.2022.12.003>
- Vendramin R., Verheyden Y., Ishikawa H., Goedert L., Nicolas E., Saraf K., Leucci E (2018) **SAMSON fosters cancer cell fitness by concertedly enhancing mitochondrial and cytosolic translation** *Nat Struct Mol Biol* **25**:1035–1046 <https://doi.org/10.1038/s41594-018-0143-4>
- Vyas A., Harbison R. A., Faden D. L., Kubik M., Palmer D., Zhang Q., Duvvuri U. (2021) **Recurrent Human Papillomavirus-Related Head and Neck Cancer Undergoes Metabolic Reprogramming and Is Driven by Oxidative Phosphorylation** *Clin Cancer Res* **27**:6250–6264 <https://doi.org/10.1158/1078-0432.ccr-20-4789>
- Wang Q., Chai D., Sobhani N., Sun N., Neeli P., Zheng J., Tian H (2022) **C1QBP regulates mitochondrial plasticity to impact tumor progression and antitumor immune response** *Front Physiol* **13** <https://doi.org/10.3389/fphys.2022.1012112>
- Wang Y., Zhang Q., Deng X., Wang Y., Tian X., Zhang S., Li J. (2024) **PA28y induces dendritic cell maturation and activates T-cell immune responses in oral lichen planus** *MedComm* **5** <https://doi.org/10.1002/mco2.561>
- Wang Z., Wu V. H., Allevato M. M., Gilardi M., He Y., Luis Callejas-Valera J., Gutkind J. S (2019) **Syngeneic animal models of tobacco-associated oral cancer reveal the activity of in situ anti-CTLA-4** *Nat Commun* **10** <https://doi.org/10.1038/s41467-019-13471-0>
- Westermann B (2010) **Mitochondrial fusion and fission in cell life and death** *Nature Reviews Molecular Cell Biology* **11**:872–884 <https://doi.org/10.1038/nrm3013>

Xia L., Oyang L., Lin J., Tan S., Han Y., Wu N., Liao Q. (2021) **The cancer metabolic reprogramming and immune response** *Mol Cancer* **20** <https://doi.org/10.1186/s12943-021-01316-8>

Xiao L., Hu Q., Peng Y., Zheng K., Zhang T., Yang L., Liu Y. (2021) **TRAP1 suppresses oral squamous cell carcinoma progression by reducing oxidative phosphorylation metabolism of Cancer-associated fibroblasts** *BMC Cancer* **21** <https://doi.org/10.1186/s12885-021-09049-z>

Xu L., Xiao N., Liu F., Ren H., Gu J (2009) **Inhibition of RIG-I and MDA5-dependent antiviral response by gC1qR at mitochondria** *Proc Natl Acad Sci USA* **106**:1530–1535 <https://doi.org/10.1073/pnas.0811029106>

Yao L., Xuan Y., Zhang H., Yang B., Ma X., Wang T., Zhang B. (2021) **Reciprocal REGγ-mTORC1 regulation promotes glycolytic metabolism in hepatocellular carcinoma** *Oncogene* **40**:677–692 <https://doi.org/10.1038/s41388-020-01558-8>

Zhang Z., Gao Z., Rajthala S., Sapkota D., Dongre H., Parajuli H., Liang X (2020) **Metabolic reprogramming of normal oral fibroblasts correlated with increased glycolytic metabolism of oral squamous cell carcinoma and precedes their activation into carcinoma associated fibroblasts** *Cell Mol Life Sci* **77**:1115–1133 <https://doi.org/10.1007/s00018-019-03209-y>

Zhu Z., Liu Y., Wu D., Wang H (2021) **Association Between Mitochondrial DNA Copy Number and Head and Neck Squamous Cell Carcinoma: A Systematic Review and Dose-Response Meta-Analysis** *Med Sci Monit* **27** <https://doi.org/10.12659/msm.928327>

Editors

Reviewing Editor

Sameh Ali

Children's Cancer Hospital Egypt, Cairo, Egypt

Senior Editor

Richard White

University of Oxford, Oxford, United Kingdom

Reviewer #1 (Public review):

Summary:

In this manuscript, the authors have tried to dissect the functions of Proteasome activator 28γ (PA28γ) which is known to activate proteasomal function in an ATP-independent manner. Although there are multiple works that have highlighted the role of this protein in tumours, this study specifically tried to develop a correlation with Complement C1q binding protein (C1QBp) that is associated with immune response and energy homeostasis.

Strengths:

The observations of the authors hint that beyond PA28γ's association with the proteasome, it might also stabilize certain proteins such as C1QBP which influences energy metabolism.

Weaknesses:

The strength of the work also becomes its main drawback. That is, how PA28y stabilizes C1QBP or how C1QBP elicits its pro-tumourigenic role under PA28y OE.

In most of the experiments, the authors have been dependent on the parallel changes in the expression of both the proteins to justify their stabilizing interaction. However, this approach is indirect at best and does not confirm the direct stabilizing effect of this interaction. IP experiments do not indicate direct interaction and have some quality issues. The upregulation of C1QBP might be indirect at best. It is quite possible that PA28y might be degrading some secondary protein/complex that is responsible for C1QBP expression. Since the core idea of the work is PA28y direct interaction with C1QBP stabilizing it, the same should be demonstrated in a more convincing manner.

In all of the assays, C1QBP has been detected as doublet. However, the expression pattern of the two bands varies depending on the experiment. In some cases, the upper band is intensely stained and in some the lower bands. Do C1QBP isoforms exist and are they differentially regulated depending on experiment conditions/tissue types?

Problems with the background of the work: Line 76. This statement is far-fetched. There are presently a number of works of literature that have dealt with the metabolic programming of OSCC including identification of specific metabolites. Moreover, beyond the estimation of OCR, the authors have not conducted any experiments related to metabolism. In the Introduction, the significance of this study and how it will extend our understanding of OSCC needs to be elaborated.

<https://doi.org/10.7554/eLife.101244.1.sa1>

Reviewer #2 (Public review):

Summary:

The authors tried to determine how PA28g functions in oral squamous cell carcinoma (OSCC) cells. They hypothesized it may act through metabolic reprogramming in the mitochondria.

Strengths:

They found that the genes of PA28g and C1QBP are in an overlapping interaction network after an analysis of a genome database. They also found that the two proteins interact in coimmunoprecipitation and pull-down assays using the lysate from OSCC cells with or without expression of the exogenous genes. They used truncated C1QBP proteins to map the interaction site to the N-terminal 167 residues of C1QBP protein. They observed the levels of the two proteins are positively correlated in the cells. They provided evidence for the colocalization of the two proteins in the mitochondria, the effect on mitochondrial form and function in vitro and in vivo OSCC models, and the correlation of the protein expression with the prognosis of cancer patients.

Weaknesses:

Many data sets are shown in figures that cannot be understood without more descriptions, either in the text or the legend, e.g., Figure 1A. Similarly, many abbreviations are not defined.

Some of the pull-down and coimmunoprecipitation data do not support the conclusion about the PA28g-C1QBP interaction. For example, in Appendix Figure 1B the Flag-C1QBP was detected in the Myc beads pull-down when the protein was expressed in the 293T cells without the Myc-PA28g, suggesting that the pull-down was not due to the interaction of the C1QBP and PA28g proteins. In Appendix Figure 1C, assume the SFB stands for a biotin tag, then the SFB-PA28g should be detected in the cells expressing this protein after pull-down by

streptavidin; however, it was not. The Western blot data in Figure 1E and many other figures must be quantified before any conclusions about the levels of proteins can be drawn.

The immunoprecipitation method is flawed as it is described. The antigen (PA28g or C1QBP) should bind to the respective antibody that in turn should binds to Protein G beads. The resulting immunocomplex should end up in the pellet fraction after centrifugation and be analyzed further by Western blot for coprecipitates. However, the method in the Appendix states that the supernatant was used for the Western blot.

To conclude that PA28g stabilizes C1QBP through their physical interaction in the cells, one must show whether a protease inhibitor can substitute PA28g and prevent C1QBP degradation, and also show whether a mutation that disrupts the PA28g-C1QBP interaction can reduce the stability of C1QBP. In Figure 1F, all cells expressed Myc-PA28g. Therefore, the conclusion that PA28g prevented C1QBP degradation cannot be reached. Instead, since more Myc-PA28g was detected in the cells expressing Flag-C1QBP compared to the cells not expressing this protein, a conclusion would be that the C1QBP stabilized the PA28g. Figure 1G is a quantification of Western blot data that should be shown.

The binding site for PA28g in C1QBP was mapped to the N-terminal 167 residues using truncated proteins. One caveat would be that some truncated proteins did not fold correctly in the absence of the sequence that was removed. Thus, the C-terminal region of the C1QBP with residues 168-283 may still bind to the PA29g in the context of full-length protein. In Figure 1I, more Flag-C1QBP 1-167 was pulled down by Myc-PA28g than the full-length protein or the Flag-C1QBP 1-213. Why?

The interaction site in PA28g for C1QBP was not mapped, which prevents further analysis of the interaction. Also, if the interaction domain can be determined, structural modeling of the complex would be feasible using AlphaFold2 or other programs. Then, it is possible to test point mutations that may disrupt the interaction and if so, the functional effect

<https://doi.org/10.7554/eLife.101244.1.sa0>